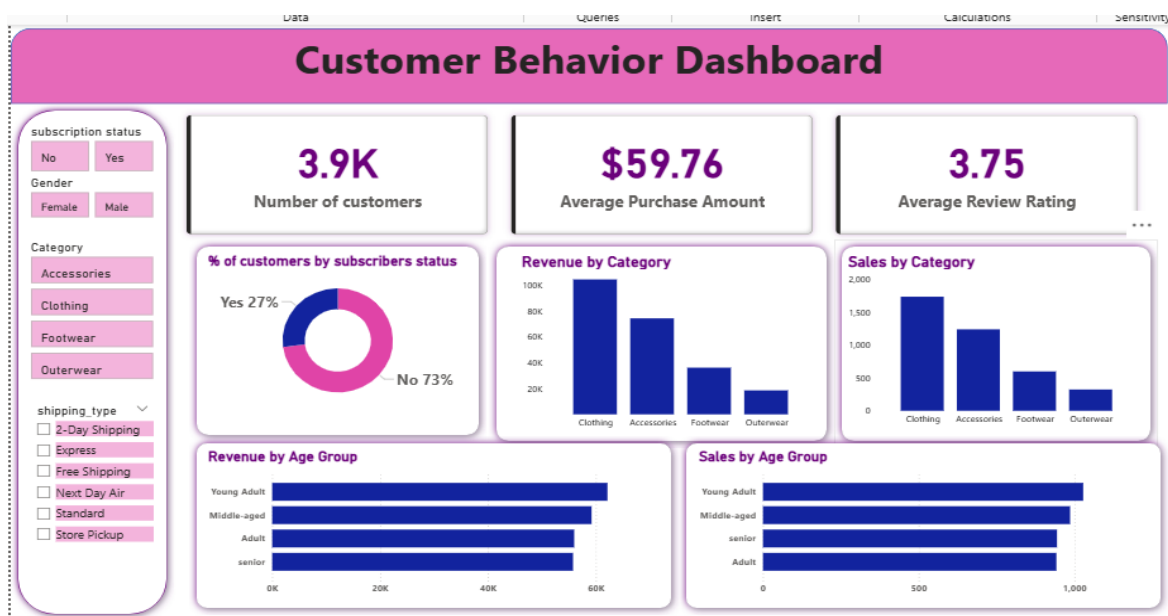


# Unlocking Retail Growth: Transforming Consumer Data into Targeted Marketing and Retention Strategies.

## 1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.



## 1.1 Introduction

In today's rapidly evolving retail landscape, understanding consumer behavior across multiple touchpoints is critical to sustaining market competitiveness. Modern consumers interact with brands through diverse distribution channels, ranging from digital storefronts to physical retail locations and their purchasing decisions are heavily influenced by promotions, product feedback, seasonal dynamics, and payment options.

While the organization collects substantial volumes of transactional data, raw data alone does not yield a competitive advantage. To unlock true commercial value, the company must transform data points into structured, actionable intelligence that

directly informs marketing tactics, inventory allocation, and customer retention programs.

## **2. Problem Statement**

### **2.1 Current Business Challenge**

A leading retail organization is experiencing shifting customer purchasing patterns across demographics, product categories, and sales channels (online vs. offline).

The retail company wants to sell more, keep customers happy, and bring them back to shop again. Today, people buy products in different ways some shop online, some go to physical stores, some wait for discounts, and others look closely at reviews before buying.

We have a lot of sales data, but right now, it isn't giving us clear answers. This project cleans up that data and builds easy-to-use visual reports so the team can make smarter business decisions.

Customers' buying habits are changing across different age groups, product types, and shopping channels. Right now, we do not clearly know:

- Do discounts really make people buy more, or do they just lower our profit?
- How much do customer reviews and payment choices (like credit card vs. cash) affect sales?
- Why do some shoppers buy once and never come back?

### **The Key Question We Need to Answer**

**“How can we use our shopping data to spot sales trends, keep customers coming back, and make better marketing and product decisions?”**

## Dataset Summary

The dataset covers **3,900 individual purchase records** across **18 columns** containing demographic, transaction, and behavioral attributes:

| Customer | Age | Gender | Item Purchased | Category    | Purchase | Location  | Size | Color     | Season | Review | R   | Subscript  | Shipping | Discount | Promo Co | Previous Purchases | Payment Method | Frequency |
|----------|-----|--------|----------------|-------------|----------|-----------|------|-----------|--------|--------|-----|------------|----------|----------|----------|--------------------|----------------|-----------|
| 1        | 55  | Male   | Blouse         | Clothing    | 53       | Kentucky  | L    | Gray      | Winter | 3.1    | Yes | Express    | Yes      | Yes      |          | 14                 | Venmo          | Fortnight |
| 2        | 19  | Male   | Sweater        | Clothing    | 64       | Maine     | L    | Maroon    | Winter | 3.1    | Yes | Express    | Yes      | Yes      |          | 2                  | Cash           | Fortnight |
| 3        | 50  | Male   | Jeans          | Clothing    | 73       | Massachu  | S    | Maroon    | Spring | 3.1    | Yes | Free Ship  | Yes      | Yes      |          | 23                 | Credit Card    | Weekly    |
| 5        | 21  | Male   | Sandals        | Footwear    | 90       | Rhode Isl | M    | Maroon    | Spring | 3.5    | Yes | Next Day   | Yes      | Yes      |          | 49                 | PayPal         | Weekly    |
| 6        | 45  | Male   | Blouse         | Clothing    | 49       | Oregon    | M    | Turquoise | Spring | 2.7    | Yes | Free Ship  | Yes      | Yes      |          | 31                 | PayPal         | Annually  |
| 7        | 46  | Male   | Sneakers       | Footwear    | 20       | Wyoming   | M    | White     | Summer | 2.9    | Yes | Standard   | Yes      | Yes      |          | 14                 | Venmo          | Weekly    |
| 8        | 63  | Male   | Shirt          | Clothing    | 85       | Montana   | M    | Gray      | Fall   | 3.2    | Yes | Free Ship  | Yes      | Yes      |          | 49                 | Cash           | Quarterly |
| 9        | 27  | Male   | Shorts         | Clothing    | 34       | Louisiana | L    | Charcoal  | Winter | 3.2    | Yes | Free Ship  | Yes      | Yes      |          | 19                 | Credit Card    | Weekly    |
| 10       | 26  | Male   | Coat           | Outerwear   | 97       | West Virg | L    | Silver    | Summer | 2.6    | Yes | Express    | Yes      | Yes      |          | 8                  | Venmo          | Annually  |
| 11       | 57  | Male   | Handbag        | Accessories | 31       | Missouri  | M    | Pink      | Spring | 4.8    | Yes | 2-Day Shi  | Yes      | Yes      |          | 4                  | Cash           | Quarterly |
| 12       | 53  | Male   | Shoes          | Footwear    | 34       | Arkansas  | L    | Purple    | Fall   | 4.1    | Yes | Store Pick | Yes      | Yes      |          | 26                 | Bank Transfer  | Bi-Weekl  |
| 13       | 30  | Male   | Shorts         | Clothing    | 68       | Hawaii    | S    | Olive     | Winter | 4.9    | Yes | Store Pick | Yes      | Yes      |          | 10                 | Bank Transfer  | Fortnight |
| 14       | 61  | Male   | Coat           | Outerwear   | 72       | Delaware  | M    | Gold      | Winter | 4.5    | Yes | Express    | Yes      | Yes      |          | 37                 | Venmo          | Fortnight |
| 15       | 65  | Male   | Dress          | Clothing    | 51       | New Ham   | M    | Violet    | Spring | 4.7    | Yes | Express    | Yes      | Yes      |          | 31                 | PayPal         | Weekly    |
| 16       | 64  | Male   | Coat           | Outerwear   | 53       | New York  | L    | Teal      | Winter | 4.7    | Yes | Free Ship  | Yes      | Yes      |          | 34                 | Debit Card     | Weekly    |
| 17       | 64  | Male   | Skirt          | Clothing    | 81       | Rhode Isl | M    | Teal      | Winter | 2.8    | Yes | Store Pick | Yes      | Yes      |          | 8                  | PayPal         | Monthly   |
| 18       | 25  | Male   | Sunglasses     | Accessories | 36       | Alabama   | S    | Gray      | Spring | 4.1    | Yes | Next Day   | Yes      | Yes      |          | 44                 | Debit Card     | Bi-Weekl  |
| 19       | 53  | Male   | Dress          | Clothing    | 38       | Mississip | XL   | Lavender  | Winter | 4.7    | Yes | 2-Day Shi  | Yes      | Yes      |          | 36                 | Venmo          | Quarterly |
| 20       | 52  | Male   | Sweater        | Clothing    | 48       | Montana   | S    | Black     | Summer | 4.6    | Yes | Free Ship  | Yes      | Yes      |          | 17                 | Cash           | Weekly    |
| 21       | 66  | Male   | Pants          | Clothing    | 90       | Rhode Isl | M    | Green     | Summer | 3.3    | Yes | Standard   | Yes      | Yes      |          | 46                 | Debit Card     | Bi-Weekl  |
| 22       | 21  | Male   | Pants          | Clothing    | 51       | Louisiana | M    | Black     | Winter | 2.8    | Yes | Express    | Yes      | Yes      |          | 50                 | Cash           | Every 3 M |
| 23       | 31  | Male   | Pants          | Clothing    | 62       | North Can | M    | Charcoal  | Winter | 4.1    | Yes | Store Pick | Yes      | Yes      |          | 22                 | Debit Card     | Quarterly |

Customer Demographics - Age, Gender, Location, Subscription Status

Purchase Details- item Purchased, Category, Purchase Amount (USD), Season, Size, Color

Shopping Behavior - Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

## Data Preparation & Cleaning Techniques (Python)

Data engineering and preprocessing were performed in Python using the pandas library to clean, validate, and prepare the dataset for analysis:

1. **Data Loading:** Imported the dataset using pandas.
2. **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics, customer ages spanned from **18 to 70 years** (average: 44.07 years) and purchase amounts ranged from **\$20 to \$100** without invalid negative numbers or extreme data entry errors

|        | Customer ID | Age         | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied |
|--------|-------------|-------------|--------|----------------|----------|-----------------------|----------|------|-------|--------|---------------|---------------------|---------------|------------------|
| count  | 3900.000000 | 3900.000000 | 3900   | 3900           | 3900     | 3900.000000           | 3900     | 3900 | 3900  | 3900   | 3863.000000   | 3900                | 3900          | 39               |
| unique | NaN         | NaN         | 2      | 25             | 4        | NaN                   | 50       | 4    | 25    | 4      | NaN           | 2                   | 6             |                  |
| top    | NaN         | NaN         | Male   | Blouse         | Clothing | NaN                   | Montana  | M    | Olive | Spring | NaN           | No                  | Free Shipping |                  |
| freq   | NaN         | NaN         | 2652   | 171            | 1737     | NaN                   | 96       | 1755 | 177   | 999    | NaN           | 2847                | 675           | 22               |
| mean   | 1950.500000 | 44.068462   | NaN    | NaN            | NaN      | 59.764359             | NaN      | NaN  | NaN   | NaN    | 3.750065      | NaN                 | NaN           | NaN              |
| std    | 1125.977353 | 15.207589   | NaN    | NaN            | NaN      | 23.685392             | NaN      | NaN  | NaN   | NaN    | 0.716983      | NaN                 | NaN           | NaN              |
| min    | 1.000000    | 18.000000   | NaN    | NaN            | NaN      | 20.000000             | NaN      | NaN  | NaN   | NaN    | 2.500000      | NaN                 | NaN           | NaN              |
| 25%    | 975.750000  | 31.000000   | NaN    | NaN            | NaN      | 39.000000             | NaN      | NaN  | NaN   | NaN    | 3.100000      | NaN                 | NaN           | NaN              |
| 50%    | 1950.500000 | 44.000000   | NaN    | NaN            | NaN      | 60.000000             | NaN      | NaN  | NaN   | NaN    | 3.800000      | NaN                 | NaN           | NaN              |
| 75%    | 2925.250000 | 57.000000   | NaN    | NaN            | NaN      | 81.000000             | NaN      | NaN  | NaN   | NaN    | 4.400000      | NaN                 | NaN           | NaN              |
| max    | 3900.000000 | 70.000000   | NaN    | NaN            | NaN      | 100.000000            | NaN      | NaN  | NaN   | NaN    | 5.000000      | NaN                 | NaN           | NaN              |

| Discount Applied | Promo Code Used | Previous Purchases | Payment Method | Frequency of Purchases |
|------------------|-----------------|--------------------|----------------|------------------------|
| 3900             | 3900            | 3900.000000        | 3900           | 3900                   |
| 2                | 2               | NaN                | 6              | 7                      |
| No               | No              | NaN                | PayPal         | Every 3 Months         |
| 2223             | 2223            | NaN                | 677            | 584                    |
| NaN              | NaN             | 25.351538          | NaN            | NaN                    |
| NaN              | NaN             | 14.447125          | NaN            | NaN                    |
| NaN              | NaN             | 1.000000           | NaN            | NaN                    |
| NaN              | NaN             | 13.000000          | NaN            | NaN                    |
| NaN              | NaN             | 25.000000          | NaN            | NaN                    |
| NaN              | NaN             | 38.000000          | NaN            | NaN                    |
| NaN              | NaN             | 50.000000          | NaN            | NaN                    |

- Completeness Audit:** Evaluated missingness across all 18 attributes using `df.isnull().sum()`. The dataset demonstrated high overall completeness, with 17 out of 18 columns being 100% complete.

4. **Targeted Missing Value Imputation:** Identified **37 missing values** exclusively in the Review Rating column (representing less than 1% of total records).
5. **Median Category Imputation:** Rather than dropping incomplete records or applying a global mean (which could distort satisfaction metrics), missing rating values were imputed using the **median rating of each corresponding product category**. This preserved statistical distribution without introducing biased extremes.
6. **Column Standardization & Cleaning:** Converted all column header names to lowercase snake\_case (e.g., Customer ID - customer\_id, Purchase Amount (USD) - purchase\_amount) to maintain consistent formatting for database loading

```
[11] df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
Python

[12] df.columns
Python
... Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount', 'location', 'size', 'color', 'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'promo_code_used', 'previous_purchases',
'payment_method', 'frequency_of_purchases'],
dtype='object')

[13] #create a column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
Python
```

## 7. Feature Engineering & Redundancy Elimination:

- ✓ **Age Binning:** Binned continuous age values into discrete categories (age\_group: *Young Adult, Middle-aged, Adult, Senior*) to enable demographic cohort analysis.
- ✓ **Buying Frequency:** Calculated purchase\_frequency\_days to track buying velocity over time.
- ✓ **Redundant Feature Removal:** Evaluated promo\_code\_used against discount\_applied. Because both columns contained identical duplicate information, promo\_code\_used was dropped to streamline the schema.

8. **Database Integration Pipeline:** Established an engine connection between Python and **PostgreSQL** to automatically upload the cleaned, fully completed dataframe into structured database tables for advanced querying.

## Data Analysis using SQL (Business Transactions)

### 1. Revenue Contribution by Gender

Male buyers account for over double the revenue of female buyers due to a higher transaction count (2,652 vs. 1,248 transactions)

- **Male Revenue:** \$157,890 (2,652 buyers)
- **Female Revenue:** \$75,191 (1,248 buyers)

|   | gender<br>text | revenue<br>numeric |
|---|----------------|--------------------|
| 1 | Female         | 75191              |
| 2 | Male           | 157890             |

### 2. High-Spending Discount Users

- **839 customers** utilized a promotional discount while still maintaining an order total higher than the overall average purchase amount of \$59.76.

|                 | customer_id<br>bigint | purchase_amount<br>bigint   |
|-----------------|-----------------------|-----------------------------|
| 1               | 2                     | 64                          |
| 2               | 3                     | 73                          |
| 3               | 4                     | 90                          |
| 4               | 7                     | 85                          |
| 5               | 9                     | 97                          |
| 6               | 12                    | 68                          |
| 7               | 13                    | 72                          |
| 8               | 16                    | 81                          |
| 9               | 20                    | 90                          |
| 10              | 22                    | 62                          |
| 11              | 24                    | 88                          |
| Total rows: 839 |                       | Query complete 00:00:00.133 |

### 3. Top 5 Rated Products

Products with the highest average customer review ratings across the catalog:

|   | item_purchased<br>text | Average Product Rating<br>numeric |
|---|------------------------|-----------------------------------|
| 1 | Gloves                 | 3.86                              |
| 2 | Sandals                | 3.84                              |
| 3 | Boots                  | 3.82                              |
| 4 | Hat                    | 3.80                              |
| 5 | Skirt                  | 3.78                              |

**4. Shipping Option Comparison-** Compared average purchase amounts between Standard and Express shipping.

|   | shipping_type<br>text | round<br>numeric |
|---|-----------------------|------------------|
| 1 | Standard              | 58.46            |
| 2 | Express               | 60.48            |

**5. Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

|   | subscription_status<br>text | total_customers<br>bigint | avg_spend<br>numeric | total_revenue<br>numeric |
|---|-----------------------------|---------------------------|----------------------|--------------------------|
| 1 | Yes                         | 1053                      | 59.49                | 62645.00                 |
| 2 | No                          | 2847                      | 59.87                | 170436.00                |

**6. Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

|   | item_purchased<br>text | discount_rate<br>numeric |
|---|------------------------|--------------------------|
| 1 | Hat                    | 50.00                    |
| 2 | Sneakers               | 49.66                    |
| 3 | Coat                   | 49.07                    |
| 4 | Sweater                | 48.17                    |
| 5 | Pants                  | 47.37                    |

**7. Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

|   | customer_segment<br>text | Number of Customers<br>bigint |
|---|--------------------------|-------------------------------|
| 1 | Loyal                    | 3116                          |
| 2 | New                      | 83                            |
| 3 | Returning                | 701                           |

**8. Top 3 Products per Category** – Listed the most purchased products within each category.

|    | item_rank<br>bigint | category<br>text | item_purchased<br>text | total_orders<br>bigint |
|----|---------------------|------------------|------------------------|------------------------|
| 1  | 1                   | Accessories      | Jewelry                | 171                    |
| 2  | 2                   | Accessories      | Sunglasses             | 161                    |
| 3  | 3                   | Accessories      | Belt                   | 161                    |
| 4  | 1                   | Clothing         | Blouse                 | 171                    |
| 5  | 2                   | Clothing         | Pants                  | 171                    |
| 6  | 3                   | Clothing         | Shirt                  | 169                    |
| 7  | 1                   | Footwear         | Sandals                | 160                    |
| 8  | 2                   | Footwear         | Shoes                  | 150                    |
| 9  | 3                   | Footwear         | Sneakers               | 145                    |
| 10 | 1                   | Outerwear        | Jacket                 | 163                    |
| 11 | 2                   | Outerwear        | Coat                   | 161                    |

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

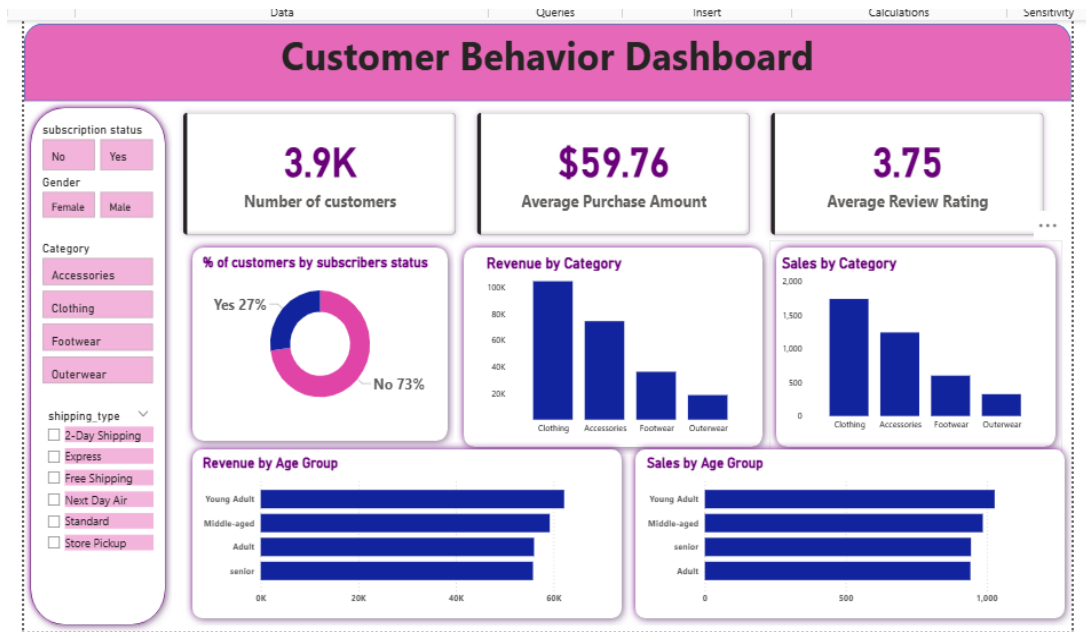
|   | subscription_status<br>text | repeat_buyers<br>bigint |
|---|-----------------------------|-------------------------|
| 1 | No                          | 2518                    |
| 2 | Yes                         | 958                     |

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group

|   | age_group<br>text | total_revenue<br>numeric |
|---|-------------------|--------------------------|
| 1 | Young Adult       | 62143                    |
| 2 | Middle-aged       | 59197                    |
| 3 | Adult             | 55978                    |
| 4 | Senior            | 55763                    |

## Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



The Power BI dashboard was built to communicate these metrics visually to stakeholders:

- **Header KPIs:** Top summary cards highlighting **3.9K Customers**, **\$59.76 Average Purchase Amount**, and **3.75 Average Review Rating**.
- **Subscription Donut Chart:** Displays **27% Yes** vs. **73% No** subscription share.
- **Category Visuals:** Bar charts breaking down total revenue and order volume across **Clothing**, **Accessories**, **Footwear**, and **Outerwear**.
- **Age Breakdown:** Horizontal bar graphs comparing spending trends across **Young Adult**, **Middle-aged**, **Adult**, and **Senior** cohorts.
- **Left Filter Panel:** Interactive slicers allowing filtering by *Subscription* Status, Gender, Category, and Shipping Type (2-Day, Express, Free Shipping, Next Day Air, Standard, Store Pickup)

### Strategic Recommendations

1. **Convert Repeat Buyers into Subscribers:** Over 2,500 repeat buyers are not enrolled in subscriptions. Offer immediate subscriber perks such as free shipping upgrades or exclusive discounts to convert this high-intent audience.
2. **Refine Discount Strategies to Protect Profit Margins:** Since half of all Hat and Sneaker purchases rely on discounts, replace sitewide price cuts with product-bundle offers (e.g., "Buy a Blouse, get a Hat at 20% off") to boost order volume without lowering item value.

3. **Expand Marketing to Female Demographic:** Female shoppers generate 32.3% of total revenue. Increase targeted campaigns focused on top-rated items like Blouses, Jewelry, and Sandals to grow this customer segment.
4. **Leverage Shipping Preferences to Boost Basket Size:** Express shipping customers spend more per order (\$60.48) than standard shipping users (\$58.46). Implement a spend threshold (e.g., "Free Express Shipping on orders over \$75") to encourage higher per-transaction spending